

Nelly Barret

ASSISTANT PROFESSOR AT INSA LYON AND LIRIS

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Research interests

I work in the broad field of data management, and more precisely my research interests focus on heterogeneous and/or multimodal data, data integration systems, and data exploration. Recently, I also added Federated Learning to my research interests. The projects that I worked on have always been part of multidisciplinary contexts, including urban planning, journalism, and health, and have been exemplified on large, complex, and often real-world data.



Academic positions

2025 - now	Assistant Professor INSA Lyon & LIRIS (FR)
2024 - 2025	Postdoctoral researcher Politecnico di Milano (IT)
2021 - 2024	PhD student Institut Polytechnique de Paris & Inria Saclay (FR)
2020 - 2020	Master intern LIRIS (FR)
2018 - 2018	Bachelor intern LIRIS (FR)



Education

2021 - 2024	PhD degree Institut Polytechnique de Paris
2018 - 2020	Master's degree Université Claude Bernard Lyon 1
2015 - 2018	Bachelor's degree Université Claude Bernard Lyon 1



Research visits

Institut de Recherche en Informatique de Toulouse (Oct. 2025 (2 days))



Awards

PhD thesis award (Oct. 2024)



Research projects

BETTER: real-world health-data distributed analytics research platform (Apr. 2024 – July 2025), CONNECTIONSTUDIO: user data exploration of heterogeneous data (March 2023 – March 2024), PATHWAYS: find interesting entity-to-entity paths in heterogeneous data (May 2022 – March 2024), ABSTRA: Entity-Relationship summaries from semi-structured data (Jan. 2021 – March 2024), PREDIHOOD: supervised prediction of the environment of neighborhoods (Feb. 2020 – July 2020), GEOALIGN: matching and merging geographic entities (Jan. 2019 – June 2019), VIZLIRIS: recommending and clustering French neighborhoods (May 2018 – July 2018),



Research software and tools

I-ETL (Apr. 2024 – July 2025), ConnectionStudio (March 2023 – March 2024), PathWays (May 2022 – March 2024), Abstra (Jan. 2021 – March 2024), Predihood (Feb. 2020 – July 2020), GeoAlign (Jan. 2019 – June 2019), VizLIRIS (May 2018 – July 2018)



Working groups

Commission égalité & parité femme-homme (Nov. 2025 – now), Diversity, Equity and Inclusion (Apr. 2023 – now), FAIRification of genomic annotations (Dec. 2024 – now)



Publications

My complete list of publications is available on my [ORCID record](#).



Professional service

Conference and workshop organizer: CORIA-TALN (Conférence en Recherche d'Information et Applications - Traitement Automatique des Langues Naturelles): 2027

Conference and workshop chair: CMLS (Conceptual Modeling for Life Sciences workshop): 2025 (workshop chair), 2026 (workshop chair), EDBT/ICDT Joint Conference (Extending DataBase Technology): 2027 (Diversity & Inclusion Chair),

Conference program committee member: ICDE (International Conference on Data Engineering): 2025 (reviewer), CAiSE (Conference on Advanced Information Systems Engineering): 2025 (reviewer), CIDR (Conference on Innovative Data systems Research): 2023 (reviewer), CIKM (International Conference on Information and Knowledge Management): 2026 (pc member), EDBT (Extending DataBase Technology): 2023 (reviewer), iSAILS (Information Systems and AI for Life Sciences workshop): 2025 (pc member), TGD (Transforming Graph Data workshop): 2026 (pc member), ICWE (International Conference on Web Engineering): 2021 (reviewer), RCIS (Research Challenges in Information Science): 2026 (forum pc member), ADBIS (Advances in Databases and Information Systems): 2026 (pc member), 2025 (pc member), DBKDA (Databases, Knowledge, and Data Applications): 2025 (pc member), BDA (Gestion de Données – Principes, Technologies et Applications): 2025 (demonstrations reviewer), 2026 (pc member),

Journal review board member: VLDB Proceedings (Proceedings of the Very Large DataBases volume): 2027

Journal reviewer: BMC Bioinformatics: 2025, BMC Medical informatics and Decision Making: 2024, GigaScience: 2025, PLoS One: 2024, Scientific Reports: 2024, JDIQ (Journal of Data and Information Quality): 2025, CMC (Computers, Materials and Continua): 2025, JOSS (Journal of Open Source Software): 2025,



Institutional responsibilities

2026 - now	GRID500 platform LIRIS referent main maintainer
2025 - now	DRIM webpage maintainer main maintainer
2024 - 2025	LinkedIn page maintainer main maintainer
2023 - 2024	PhD representative main representative
2021 - 2024	Team seminar organizer main organizer
2021 - 2021	Internship application website contributor active contributor



Talks

My complete list of talks on my [webpage](#).



Teaching responsibilities

Recurring courses: ISN1: Informatique et Société Numérique (fall, engineer 1st year): algorithmics, Python development ISN2: Informatique et Société Numérique (spring, engineer 1st year): algorithmics, functional decomposition, introduction to networks, guided hands-on project ISN4: Informatique et Société Numérique (spring, engineer 2nd year): object-oriented programming, graphical interfaces, code versioning, hand-on project SOL: Systèmes et Outils Logiciels (fall, engineer 1st year): Microsoft Office and LibreOffice suite, basic Linux commands

Involvement in INSA teaching teams: ISN1, ISN2, ISN4, SOL: course improvement and/or redesign ISN1, ISN2, ISN4: Python and Visual Studio Code setup CGA: member of the Coordination and Management Committee (Comité de Gestion et d'Animation) First year admission interviews for INSA Lyon

Previous courses: CSE203: CS project (spring 2024, Ecole Polytechnique, bachelor 2nd year, in English) CSE204: Machine Learning (spring 2023, Ecole Polytechnique, bachelor 2nd year, in English) INF411: bases de programmation et d'algorithmique (fall 2022, Ecole Polytechnique, engineer 2nd year, in French) INF371: mécanismes de la programmation orientée objet (spring 2022, Ecole Polytechnique, engineer 1st year, in French)



Training



Advising

Engineer: Côme Périn on "Apprentissage multimodal au service de la santé des conducteurs" (April-Dec. 2026, ENTPE Vaulx-en-Velin)

Master students: Inès Boudjeli on "Débruitage et sélection automatique de signaux cardiaques pour la prédiction médicale" (spring 2026, Université de Dijon, master 2nd year) Ahmed Bel Hadj Youssef on "Construction automatique de domaines dans les data meshes" (fall 2025, INSA Lyon, engineer 5th year) Antoine Gauquier on "Path exploration of ConnectionLens graphs" (summer 2022, IMT Nord-Europe, master 1st year)

Bachelor students: Shay Pripstein on "Target data acquisition in open repositories" (winter 2023, École Polytechnique, bachelor 3rd year) Tudor Enache on "From simple to property graphs" (winter 2023, École Polytechnique, bachelor 3rd year) Nikola Dobricic on "Graph queries for abstractions" (summer 2023, Ecole Polytechnique, bachelor 2nd year) Jia-Jean Law on "Optimization of entity-to-entity data paths" (winter 2022, Ecole Polytechnique, bachelor 3rd year)